

MEMORANDUM

TO: MSAC512 Assessment Board & Lead Actuarial Examiner

FROM: Trainee Data Analyst

DATE: 23 September 2026

SUBJECT: Data Collection Governance, Web Ethics Audit, and Data Harmonization Pipeline

SECTION 1: DATA COLLECTION & ETHICS COMPLIANCE AUDIT

1.1 robots.txt & Terms of Use Inspection

Prior to automated data collection on https://www.property.co.zw, an operational audit of the site's robots.txt file and Terms of Use was performed to ensure legal compliance and system safety.

Disallowed Routes & Crawl Boundaries

The host site explicitly prohibits automated user-agents (User-agent: *) from accessing private endpoints and parameterized query paths. To comply with these directives, the scraper implemented strict route filtering:

1. User Session & Private Endpoints Avoided:

The crawler strictly avoided /crm/, /account/, /favourites/, /alerts/, /requests/, /compare/, /applications/, /listings/, and /myproperties/. These paths contain private portal data, user state variables, or computational CRM tools that are non-public and protected.

2. Prohibited Query Parameters Avoided:

The host disallows crawling under dynamic parameter patterns, including but not limited to:

- **Price/Spec Filters:** /*?*minprice=*, /*?*maxprice=*, /*?*minbeds=*, /*?*bathrooms=*, /*?*size=*
- **Geographic/System Identifiers:** /*?*suburb_id=*, /*?*town_id=*, /*?*province_id=*, /*?*site_id=*
- **Tracking & Analytics Query Strings:** /*?*utm_source=*, /*?*utm_medium=*, /*?*fbclid=*

3. Technical Execution Strategy:

To prevent generating recursive crawler loops or violating parameter restrictions, the scraper bypassed form-based filter submissions. Crawling was constrained to canonical index paths (e.g., /houses-for-sale/harare/harare-west/madokero) and listing endpoints specified within the site's public sitemap index (/content/sitemap/advert-en.xml).

1.2 Rate-Limiting Protocol & Justification

- **Enforced Delays:**
 - **HTML Index & Detail Requests:** 2.0 seconds inter-request pause (REQUEST_DELAY_SECONDS = 2.0).
 - **Media/Photograph Downloads:** 0.3 seconds per asset.
- **Engineering Justification:**

At a rate of one HTML fetch every two seconds, the scraper caps out at 30 requests per minute. This operational throughput mirrors standard human navigation speeds, remaining significantly below default Web Application Firewall (WAF) throttling thresholds (typically set at ≥ 60 requests/minute). This prevents resource degradation on the target web server.

1.3 Professional Conduct & Commercial Engagement Considerations

Actuarial and data science professional codes of conduct state that technical accessibility does not grant implied legal or ethical rights to collect, use, or republish data. In a formal commercial engagement, the following frameworks apply:

- **Licensing & Terms Agreements:** Prior written consent or a commercial API syndication agreement must be executed with the platform owner before running bulk collection pipelines.
- **Intellectual Property Rights:** Textual property descriptions, property features, and photograph assets are the intellectual property of listing brokerages and property owners. Unlicensed commercial republication constitutes copyright infringement.

- **Data Privacy (Zimbabwe Cyber and Data Protection Act / GDPR):** Personal identifiers—such as individual broker phone numbers, direct email addresses, and specific property owner details—must be scrubbed or anonymized prior to model development or report distribution.
- **Takedown & Right-to-be-Forgotten Protocols:** A structured pipeline must be maintained to immediately purge records upon request from listing agencies or property owners.

SECTION 2: DATA CLEANING & HARMONIZATION METHODOLOGY

2.1 Currency Standardization (ZiG to USD)

Listings denominated in Zimbabwe Gold (ZiG) were harmonized to a standard United States Dollar (USD) base using the Reserve Bank of Zimbabwe (RBZ) interbank reference exchange rate (1 ZiG = \$0.038 USD, September 2026 reference).

$$\text{Price}_{\text{USD}} = \text{Price}_{\text{ZiG}} \times 0.038$$

2.2 Deduplication Framework

Re-listed properties cross-posted across multiple real estate agencies were identified and deduplicated using a composite key:

$$\text{Composite Key} = \{\text{Suburb}, \text{Bedrooms}, \text{Stand Size (m}^2\text{)}, \text{Standardized Price (USD)}\}$$

Any record matching an existing composite key or an identical listing URL was removed, retaining only the first observed instance.

2.3 Outlier & Data-Entry Error Treatment

To prevent model distortion from incorrect manual data entries:

1. **Price Boundaries:** Records with prices < \$5,000 (rental entries misplaced in sales sections) or > \$5,000,000 were flagged as anomalies and removed from the target set.
2. **Physical Constraints:** Stand sizes < 50 m² or > 100,000 m² (10 hectares) were set to NaN.

3. **Room Limits:** Bedroom or bathroom counts exceeding 15 were treated as data-entry errors and set to NaN.
4. **Logical Consistency Check:** Any listing where Stand Size (m^2) < Floor Area (m^2) had its floor_area_m2 reset to NaN due to physical impossibility.

2.4 Documented Missing-Value Strategy

To preserve sample size for downstream modeling, rows were **not** dropped wholesale due to missing attribute fields.

Field	Imputation / Treatment Method	Rationale
price_standardized_usd	Row Removal (Complete Case)	Target variable for prediction; cannot be imputed.
suburb	Fallback to city_town or 'Unknown Suburb'	Retains spatial categorization where possible.
bedrooms / bathrooms	Suburb-Grouped Median Imputation	Preserves overall sample size while matching local density norms.
stand_size_m2	Suburb-Grouped Median Imputation	Fills missing land sizes using regional spatial medians.
floor_area_m2	Suburb Median + Binary Missing Flag	Explicitly models structural missingness patterns for ML algorithms.